

TrES-3b

R-2715

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_220726 · created 2026-09-01T01:35:38+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459786.8382 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.056
Transit depth (Rp/Rs)^2	3.85 +/- 2.21 [%]
Orbital Inclination (inc)	82.05 +/- 2.82
Airmass coefficient 1 (a1)	0.99 +/- 0.088
Airmass coefficient 2 (a2)	0.03 +/- 0.11
Scatter in the residuals of the lightcurve fit is	8.98 %
Best Comparison Star	#2 - [224, 410]
Optimal Aperture	1.46
Optimal Annulus	14.64
Transit Duration (day)	0.054 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.056 vs 0.16309 (deviation 0.52σ)
Duration fit vs published	0.054 d vs 0.0567 d (deviation 0.09σ)
Mid-time O–C	-0.4 min
Depth SNR	3.1
Residual scatter	8.98 %

Light curve

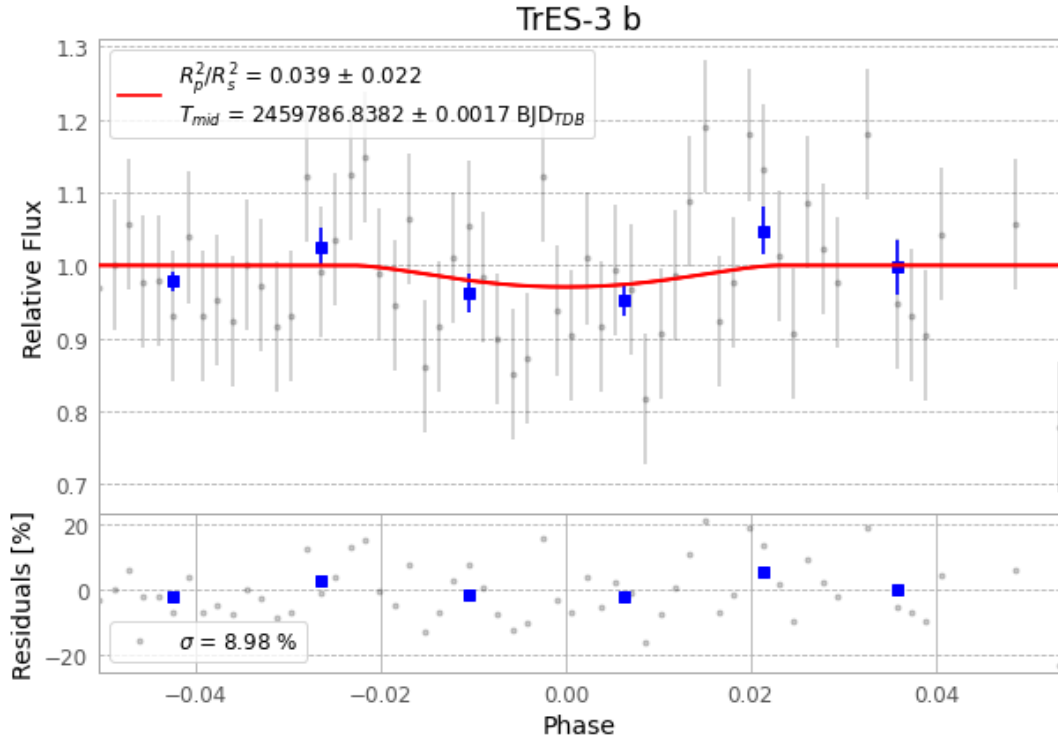


Figure 1 — FinalLightCurve_TrES-3 b_2022-07-25.png (SHA-256 575d970805549520...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [295, 326], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0a6f36b1e279c07c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

67 FITS frames (44 MB), TRES-3220726062713.FITS → TRES-3220726094512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-220726.log (9a17d27e1c9a...), FinalLightCurve_TrES-3 b_2022-07-25.png (575d97080554...), FinalLightCurve_TrES-3 b_2022-07-25.csv (9636aa6ece92...)

Compute resources

Wall time 389.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 01:35Z.